SWIFT SHIP TRACKER

Salesforce CRM Based Parcel Delivery Management System

Prepared for SkillWallet – Salesforce Developer Group Project

Platform: Salesforce Developer Edition

Project Domain: Logistics and Parcel Delivery

Core Technologies: Salesforce CRM, Flow Builder, Apex, Prompt Builder, Agentforce, Experience Cloud, Reports and Dashboards

Project Title: Swift Ship Tracker

A centralized CRM solution for booking, tracking and managing parcel deliveries

ABSTRACT

Swift Ship Tracker is a Salesforce-based customer relationship management solution designed to organize and simplify parcel delivery operations. The system addresses common operational problems such as fragmented booking information, manual status updates, limited customer visibility, repeated communication, and difficulty monitoring delivery performance. The proposed solution centralizes customer, parcel, shipment, delivery-agent, tracking and operational information inside Salesforce so that different users can work from a common source of information.

The solution supports the parcel lifecycle from booking to final delivery confirmation. A parcel can be registered with an identifier, customer information, weight, shipment status and estimated delivery date. As the shipment progresses, operational users can update the status and the system can maintain tracking information. Salesforce Flow can be used for repeatable business automation, while Apex can support business rules that require programmatic processing. Email notifications can keep customers informed when important delivery milestones occur.

The project also introduces AI-oriented capabilities. Prompt Builder can retrieve structured parcel information and present it in a consistent tracking format. Agentforce can provide a conversational interface through which users can ask shipment-related questions. Experience Cloud can provide a customer-facing self-service portal, while reports and dashboards help management understand shipment volume, delivery status, delays and agent activity.

The overall design focuses on scalability, security, automation and a better customer experience. Role-based access can separate administrative, operational and customer responsibilities. The final solution therefore demonstrates how Salesforce can be used not only as a CRM platform but also as an integrated logistics management environment.

TABLE OF CONTENTS

No.	Section	Page
1	Introduction and Project Background	4
2	Problem Statement	5
3	Project Scope and Objectives	6
4	Requirement Analysis	7
5	System Architecture	8
6	Salesforce Data Model	9
7	Customer and Shipment Management	10
8	Tracking and Delivery Lifecycle	11
9	Automation using Salesforce Flow	12
10	Apex and Programmatic Logic	13
11	Prompt Builder and AI	14
12	Agentforce Conversational Support	15
13	Security and Access Control	16
14	Experience Cloud and User Experience	17
15	Reports and Dashboards	18
16	Testing and Validation	19
17	Implementation Evidence and Conclusion	20

Note: Page numbers are arranged for this 20-page project-report format. Content can be expanded or adjusted after final institutional formatting.

1. INTRODUCTION AND PROJECT BACKGROUND

Parcel delivery has become an important part of modern commerce, education, business communication and personal logistics. As shipment volumes increase, organizations require a reliable system for registering parcels, assigning delivery operations, tracking movement and communicating with customers. A process based mainly on spreadsheets, manual calls or disconnected applications can make it difficult to maintain a consistent view of shipment status.

Swift Ship Tracker addresses this requirement by using Salesforce as a centralized platform. Salesforce provides data management, workflow automation, security, analytics and extensibility in one environment. Instead of maintaining separate systems for customer information, shipment status and operational reporting, the proposed system organizes these activities around a common CRM data model.

The project is designed around three major user groups: administrators, delivery agents and customers. Administrators manage the overall system and monitor business activity. Delivery agents work with assigned parcels and update operational statuses. Customers need a simple way to view their shipment information and receive relevant delivery updates.

Project Vision

The vision of Swift Ship Tracker is to create a connected parcel-management experience in which every shipment has a clear identity, an understandable lifecycle and an accessible status. Automation reduces repetitive work, AI improves conversational access to information, and dashboards provide management with a high-level view of operations.

Key Characteristics

- Centralized parcel and customer information.
- Structured shipment lifecycle and status management.
- Automation for repetitive operational activities.
- AI-assisted retrieval and conversational support.
- Role-based access and controlled customer visibility.
- Reports and dashboards for operational decision support.

2. PROBLEM STATEMENT

Parcel businesses frequently handle several interconnected activities: booking, pickup, dispatch, hub processing, transportation, delivery and customer communication. When these activities are recorded manually or across disconnected systems, operational teams may spend significant time checking information, updating records and responding to repetitive customer questions.

Existing Challenges

- Customers may not have a single, convenient location for tracking shipment status.
- Agents may need to switch between records or communication channels to obtain parcel information.
- Manual status updates can create inconsistent or delayed information.
- Delivery notifications may depend on employees remembering to send them.
- Managers may lack a consolidated dashboard for shipment and agent performance.
- Sensitive customer and shipment information requires controlled access.
- Operational history can be difficult to analyze if tracking events are not structured.

Proposed Solution

Swift Ship Tracker creates a centralized Salesforce solution where parcel records, customer information, delivery operations and tracking information are managed within a common platform. Automation is introduced for predictable processes, while AI capabilities provide a natural-language interface for parcel-related questions. The system is designed so that the customer experience and internal operational workflow are connected rather than isolated.

Expected Business Value

The proposed system can reduce repetitive administrative work, improve visibility, standardize operational processes and make shipment information easier to access. It also creates a structured data foundation for future enhancements such as advanced analytics, mobile delivery operations, route intelligence and additional AI capabilities.

3. PROJECT SCOPE AND OBJECTIVES

Project Scope

The project covers the design and implementation of a scalable CRM solution for parcel delivery businesses. It includes customer booking, shipment management, delivery operations, tracking, notifications, AI-powered assistance, security, customer self-service and reporting.

- Customer booking and parcel registration.
- Shipment identification using a tracking/parcel identifier.
- Parcel attributes including weight, status and estimated delivery date.
- Tracking event and delivery-status management.
- Delivery-agent operational workflow.
- Flow-based automation and email notifications.
- Apex support for complex business logic.
- Prompt Builder for structured parcel information responses.
- Agentforce for conversational support.
- Experience Cloud for customer self-service.
- Role-based access using Salesforce security features.
- Reports and dashboards for management visibility.

Objectives

1. Streamline parcel operations by automating booking, tracking and delivery notifications.
2. Improve customer satisfaction through transparent shipment status and self-service tracking.
3. Enhance agent efficiency through centralized records and guided operational actions.
4. Provide business insights using shipment, delivery and agent reports.
5. Protect customer and operational data through appropriate Salesforce security controls.

Out of Scope

Physical GPS hardware, carrier integrations, payment processing, warehouse robotics and production-scale route optimization are outside the core scope of this academic project. They can be considered future extensions after the CRM foundation is established.

4. REQUIREMENT ANALYSIS

Functional Requirements

Requirement	Description	Primary User
Customer Management	Maintain customer identity and contact information.	Admin
Parcel Booking	Create a parcel/shipment with required details.	Customer / Agent
Tracking	View current shipment status and tracking information.	Customer / Agent
Status Update	Move a parcel through operational stages.	Agent
Delivery Confirmation	Record final delivery information.	Agent
Notification	Send relevant shipment and delivery updates.	System
AI Tracking	Generate structured tracking responses.	Customer / Agent
Reporting	Monitor shipment and delivery metrics.	Admin / Manager

Non-Functional Requirements

- Scalability: the data model should support increasing parcel volume.
- Security: users should only access records appropriate to their role.
- Usability: operational actions should be simple and understandable.
- Reliability: automated processes should be predictable and testable.
- Maintainability: configuration should prefer standard Salesforce capabilities where appropriate.
- Auditability: important status changes and delivery events should be traceable.

User Stories

- As a customer, I want to book a parcel so that I can initiate a shipment.
- As a customer, I want to track my parcel so that I know its current status.
- As a delivery agent, I want to see assigned parcels so that I can complete deliveries efficiently.
- As a delivery agent, I want to update parcel status so that customers receive current information.
- As an administrator, I want dashboards so that I can monitor delivery operations.
- As a customer, I want conversational support so that I can ask shipment questions without searching multiple screens.

5. SYSTEM ARCHITECTURE

The Swift Ship Tracker architecture uses Salesforce as the central application and data platform. User interactions are handled through Salesforce interfaces, Experience Cloud and conversational AI. Business records are stored in Salesforce objects. Automation is implemented through Flow and, where appropriate, Apex. Prompt Builder and Agentforce provide AI-assisted information access. Reporting components consume structured Salesforce data.

High-Level Architecture

Customer / Delivery Agent / Administrator → Salesforce User Interface or Experience Cloud → Salesforce Data Layer → Flow / Apex Automation → Notifications / Tracking Events → Prompt Builder / Agentforce → Reports and Dashboards.

Logical Layers

Layer	Main Elements	Purpose
Presentation	Lightning UI, Experience Cloud	User interaction and self-service
Application	Flows, Apex, validation	Business processing and automation
AI	Prompt Builder, Agentforce	Conversational and generated responses
Data	Parcel, Customer, Tracking data	Centralized information storage
Security	Profiles, Permission Sets, Sharing	Controlled access
Analytics	Reports, Dashboards	Operational visibility

Scalability Approach

The solution is designed around structured records rather than hard-coded shipment information. New parcel records can follow the same data model and automation rules. Additional agents, customers and shipments can therefore be introduced without redesigning the core business process. Future integrations can consume Salesforce data through appropriate APIs or integration services.

6. SALESFORCE DATA MODEL

The data model is the foundation of Swift Ship Tracker. The primary parcel/shipment record stores information required to identify and monitor a delivery. Supporting records can represent customers, agents, tracking events and delivery confirmation. Relationships between records provide traceability from a shipment to its operational history.

Core Parcel Fields

Field	Purpose	Example
Name	Human-readable parcel name.	Chennai to Bangalore Parcel
Parcel ID	Unique business identifier.	SST-00001
Status	Current shipment lifecycle state.	In Transit
Weight	Parcel weight for operational use.	2.0 KG
Estimated Delivery Date	Expected delivery date.	05-Oct-2026
Customer	Shipment owner/requesting customer.	Sample Customer
Delivery Agent	Assigned operational user.	Agent A
Pickup Date	Date the parcel was collected.	03-Oct-2026
Delivery Date	Actual completion date.	05-Oct-2026

Tracking Event Data

A tracking event represents a meaningful change in a shipment's journey. It can include the parcel reference, status, location, timestamp and optional remarks. Maintaining tracking events separately from the current shipment status allows the system to preserve a history instead of overwriting all previous information.

Relationship Concept

A customer can be associated with multiple shipments. A shipment can have multiple tracking events and one current delivery assignment. A delivery confirmation can be associated with the shipment when the final delivery step is completed. This structure supports both current-state reporting and historical analysis.

7. CUSTOMER AND SHIPMENT MANAGEMENT

The booking workflow begins when a customer or authorized operational user creates a shipment. Required information is collected before the shipment becomes active. A tracking identifier provides a simple reference that can be shared with the customer and used for future searches.

Booking Workflow

6. Open the parcel/shipment creation interface.
7. Enter customer and receiver information.
8. Enter origin and destination information where applicable.
9. Enter parcel weight and other required attributes.
10. Create or assign a unique parcel/tracking identifier.
11. Set the initial shipment status to Booked.
12. Save the record and trigger confirmation automation.
13. Make the shipment available to authorized operational users.

Customer Experience

The customer should not need to understand Salesforce data structures. The experience should be based on simple concepts such as My Shipments, Track Parcel, Current Status, Expected Delivery and Delivery Confirmation. This abstraction makes the system easier to use and creates a consistent customer journey.

Operational Benefits

- Every parcel has a clear identifier.
- Shipment information is stored centrally.
- Agents can locate records using a known tracking reference.
- Automation reduces repetitive notification work.
- Historical tracking can support customer-service questions.
- Reports can aggregate shipment records for management.

8. TRACKING AND DELIVERY LIFECYCLE

Swift Ship Tracker follows a structured parcel lifecycle. A controlled status model helps ensure that operational users understand the current state of a shipment and that customers receive consistent information.

Shipment Lifecycle

Stage	Meaning	Typical Action
Booked	Shipment has been registered.	Booking confirmation
Picked Up	Parcel has been collected.	Pickup recorded
Dispatched	Parcel has left the origin operation.	Dispatch update
In Transit	Parcel is moving between locations.	Tracking update
Out for Delivery	Parcel is with the delivery operation.	Delivery notification
Delivered	Parcel has been successfully delivered.	Delivery confirmation

Tracking Event Concept

Whenever an important operational transition occurs, a tracking event can be recorded. The event provides historical evidence of the shipment journey. This approach is useful when a customer asks when a parcel changed status or when management needs to investigate delays.

Delivery Confirmation

When a delivery agent completes the shipment, the final status becomes Delivered. The system can record delivery date, delivery agent and optional remarks. A confirmation notification can then be sent to the customer. This final event closes the operational lifecycle of the parcel.

Exception Handling

A production version can extend the status model with exceptions such as Delayed, Address Issue, Customer Unavailable or Return to Sender. These statuses can be introduced only after defining appropriate business rules and notifications.

9. AUTOMATION USING SALESFORCE FLOW

Automation is a central part of the Swift Ship Tracker design. Salesforce Flow is appropriate for many repeatable business processes because it can respond to record changes and perform configured actions without requiring custom code for every scenario.

Flow 1 – Shipment Creation

When a new shipment is created, the flow can validate required information, assign or confirm the initial status, create a tracking event and trigger a customer confirmation email. The exact implementation depends on the final Salesforce object and field configuration.

Flow 2 – Shipment Status Update

When an agent changes the shipment status, the flow can detect the change and create a tracking-event record. It can then evaluate the new status and send an appropriate notification. For example, a transition to Out for Delivery can trigger a customer notification.

Flow 3 – Delivery Completion

When a parcel becomes Delivered, automation can set the delivery date if required, create a delivery confirmation record and send a completion notification. The process should also include conditions that prevent duplicate confirmation actions.

Automation Design Principles

- Use clear entry criteria.
- Keep each flow focused on a well-defined business process.
- Prevent repeated notifications.
- Handle missing or invalid data gracefully.
- Test both positive and negative scenarios.
- Document important flow decisions for maintainability.

Email Notification Examples

- Booking confirmation.
- Pickup confirmation.
- In-transit update.
- Out-for-delivery alert.
- Delivery completion message.

10. APEX AND PROGRAMMATIC LOGIC

Salesforce Flow handles many configuration-driven processes, but some business requirements may require Apex. Apex is used selectively in Swift Ship Tracker where programmatic logic provides value that is difficult or inappropriate to implement only through configuration.

Potential Apex Responsibilities

- Complex tracking-number generation logic.
- Bulk-safe processing of shipment records.
- Advanced validation across related records.
- Specialized integration logic.
- Reusable server-side business services.

Apex Quality Requirements

Apex logic should be bulk-safe, readable and testable. Queries and data operations should be designed to work with multiple records rather than assuming only one record is processed. Test classes should cover normal operation, invalid data and important edge cases.

Flow vs Apex

Requirement Type	Preferred Approach	Reason
Simple status update	Flow	Configuration is easier to maintain.
Email notification	Flow / Email Alert	Standard automation capability.
Create tracking event	Flow	Straightforward record automation.
Complex multi-record calculation	Apex	Programmatic processing may be clearer.
Specialized integration	Apex / Integration	Requires custom processing.
Conversational response	Prompt Builder / Agentforce	AI-specific requirement.

Testing Approach

Any Apex introduced into the project should be tested before being treated as production-ready. Tests should verify expected outcomes, handle multiple records and confirm that invalid conditions do not create corrupted shipment information.

11. PROMPT BUILDER AND AI

The Swift Ship Tracker project includes an AI-oriented Prompt Builder template called “Retrieve Parcel Details”. The purpose of this template is to retrieve structured information from a parcel record and return it in a consistent tracking-update format. This creates a bridge between Salesforce data and a natural-language response.

Prompt Objective

The prompt is designed to retrieve parcel details where the parcel identifier matches the user's input. The response format includes the parcel name, parcel ID, status, weight and estimated delivery date. A consistent format helps the downstream conversational experience provide predictable information.

Configured Data

- Parcel Name
- Parcel ID
- Status
- Weight
- Estimated Delivery Date

Prompt Builder Configuration Evidence

The supplied Salesforce screenshot shows the Prompt Builder workspace for the “Retrieve Parcel Details” template. It demonstrates the configured prompt structure and the use of parcel-related resources. The screenshot is included below as implementation evidence.

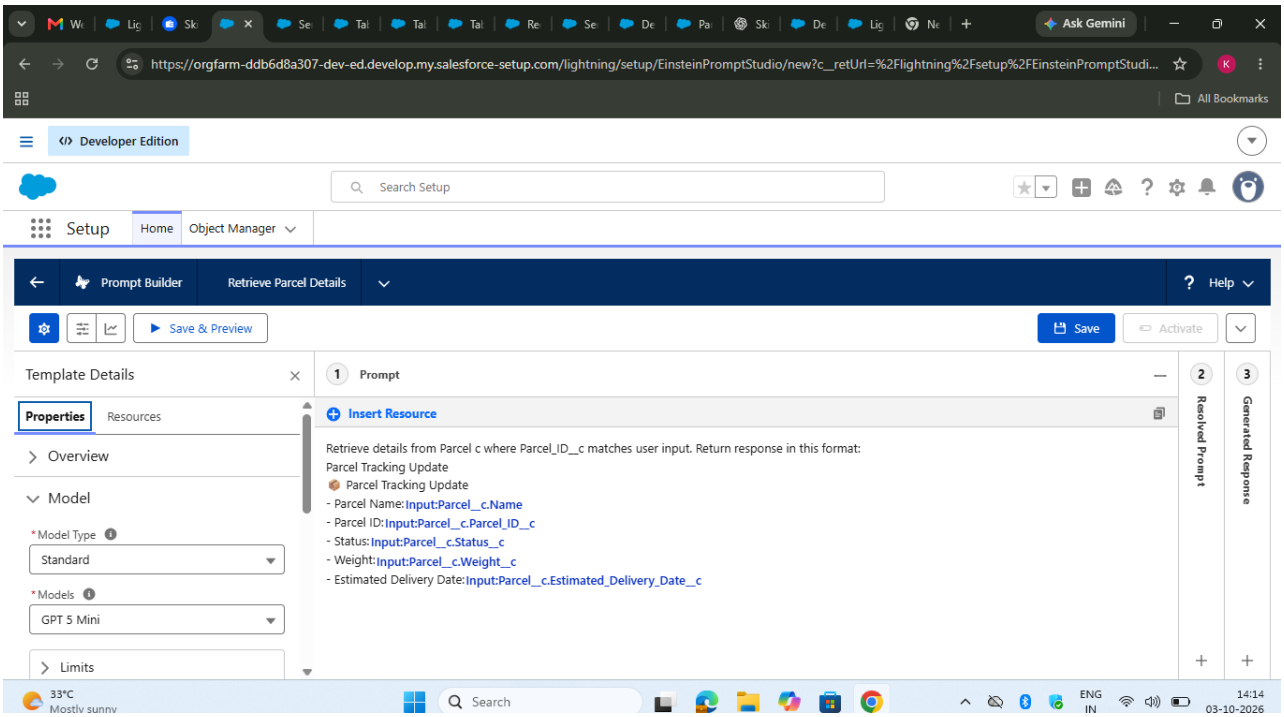


Figure 2. Prompt Builder – Retrieve Parcel Details configuration.

12. AGENTFORCE CONVERSATIONAL SUPPORT

Agentforce can provide a conversational layer over the Swift Ship Tracker data and business processes. Instead of requiring a user to navigate through several screens, an agent can ask a natural-language question such as “Where is parcel SST-00001?” and receive a response based on authorized Salesforce information.

Example Conversation

Customer: “Where is my parcel SST-00001?”

System: “The parcel is currently In Transit. The estimated delivery date is 05-Oct-2026.”

Customer: “What is the parcel weight?”

System: “The recorded parcel weight is 2.0 KG.”

AI Workflow

14. Receive the user's question.
15. Identify the shipment or parcel reference.
16. Retrieve the relevant Salesforce information.
17. Apply the configured Prompt Builder response format.
18. Return the response through the conversational interface.
19. Respect user access permissions before exposing information.

Responsible AI Considerations

- Only authorized information should be exposed.
- Responses should be grounded in current Salesforce records.
- Users should be able to distinguish system data from assumptions.
- Missing data should not be invented.
- Sensitive customer information should be protected.

Future AI Enhancements

Future versions could support delivery-delay explanations, FAQ assistance, agent guidance, automated case creation and summaries of shipment history. These enhancements should be introduced with appropriate access control and validation.

13. SECURITY AND ACCESS CONTROL

Because Swift Ship Tracker stores customer and shipment information, access control is an essential part of the design. Salesforce security features can be combined to define what a user can do and which records they can access.

Role Model

Role	Typical Access
Administrator	System configuration, all operational records and reporting.
Delivery Agent	Assigned shipment information and permitted status updates.
Customer	Own profile and authorized shipment/tracking information.

Security Mechanisms

- Profiles define baseline permissions.
- Permission Sets can provide additional permissions without creating many profiles.
- Role Hierarchy can support management visibility where appropriate.
- Sharing Rules can extend record access when business requirements require it.
- Field-level security can protect sensitive fields.
- Experience Cloud access must be configured so customers cannot view unrelated records.

Security Validation

Security testing should include attempts to access records belonging to another customer, restricted fields and unauthorized operational actions. A successful implementation should ensure that a customer cannot simply change a URL or identifier and gain access to another customer's shipment.

Auditability

Tracking events and relevant record history can provide operational evidence. Administrative monitoring and Salesforce audit capabilities can support investigation when changes need to be reviewed.

14. EXPERIENCE CLOUD AND USER EXPERIENCE

Experience Cloud can provide a customer-facing portal for Swift Ship Tracker. The purpose is to expose only the customer functionality required for parcel management rather than presenting the full internal Salesforce environment.

Customer Portal Functions

- Customer login.
- My Shipments view.
- Tracking-number search.
- Current shipment status.
- Expected delivery date.
- Tracking history where authorized.
- Delivery confirmation information.
- Support or conversational assistance.

Agent Experience

Delivery agents require a faster operational interface. A Lightning-based workspace can provide assigned shipment lists, status information and quick actions for updating delivery progress. Guided flows can reduce the number of manual steps required for common activities.

Usability Principles

- Use clear status labels.
- Keep important tracking information visible.
- Minimize unnecessary navigation.
- Use validation before saving important updates.
- Provide meaningful confirmation messages.
- Make error messages understandable to non-technical users.

Accessibility and Responsiveness

The interface should be usable across common desktop and mobile screen sizes. Labels, navigation, buttons and status indicators should remain understandable. A customer should be able to identify the current parcel status without needing technical knowledge of Salesforce.

15. REPORTS AND DASHBOARDS

Reports and dashboards convert operational records into management information. Swift Ship Tracker can use Salesforce reporting capabilities to provide a consolidated view of shipment activity, delivery performance and agent workload.

Recommended Reports

Report	Purpose
Shipment Status Report	Shows shipment counts by current status.
Daily Booking Report	Tracks parcel bookings over time.
Delayed Shipment Report	Identifies shipments requiring attention.
Agent Performance Report	Shows delivery activity by agent.
Delivery Completion Report	Measures completed deliveries.
Shipment Trend Report	Shows changes in shipment volume over time.

Dashboard Components

- Total shipments KPI.
- Booked shipment count.
- In-transit shipment count.
- Out-for-delivery count.
- Delivered shipment count.
- Delayed shipment count.
- Daily shipment trend chart.
- Agent activity chart.

Management Use

A manager can use the dashboard to identify changes in shipment volume, monitor outstanding deliveries and understand agent workload. A dashboard should not merely display numbers; each metric should correspond to a business question that the organization wants to answer.

Data Quality

Reports are only as reliable as the records behind them. Required fields, controlled status values and consistent tracking events improve the quality of reporting. Automation should also reduce the possibility of users forgetting to update related information.

16. TESTING AND VALIDATION

Testing verifies that Swift Ship Tracker behaves correctly across the parcel lifecycle and that automation, security and AI features work as intended. Testing should be performed using realistic sample records and multiple user roles.

Functional Test Cases

Test	Expected Result
Create parcel	Parcel record is created with required information.
Assign tracking ID	Tracking identifier is available for future lookup.
Update status	Current status changes and tracking history is recorded where configured.
Out for Delivery	Relevant notification automation executes.
Mark Delivered	Delivery information is recorded and confirmation is triggered.
Prompt Preview	Prompt returns parcel details using authorized record data.
Unauthorized access	User cannot view restricted shipment information.
Dashboard refresh	Reports reflect available shipment data.

Prompt Builder Validation

The Prompt Builder template should be tested with a valid parcel record and a record containing missing optional information. The response should remain structured and should not invent values. Resource references must use the correct Salesforce prompt-resource configuration rather than manually typed invalid merge syntax.

Negative Testing

- Invalid parcel identifier.
- Missing parcel record.
- Missing estimated delivery date.
- Unauthorized customer attempting to access another shipment.
- Duplicate delivery update.
- Invalid status transition.
- Multiple records processed by automation.

Final Validation Checklist

- Data model verified.
- Automation tested.
- Apex tested where used.
- AI prompt preview tested.
- Agentforce behavior tested where configured.
- Security permissions reviewed.
- Experience Cloud access reviewed.
- Reports and dashboards verified.
- GitHub/project metadata prepared.

- SkillWallet stories checked for completion.

17. IMPLEMENTATION EVIDENCE, CONCLUSION AND FUTURE WORK

Implementation Evidence

The project documentation includes the supplied Salesforce Prompt Builder screenshot as evidence of the AI configuration work. The screen shows the “Retrieve Parcel Details” template and the parcel-related information intended for the generated tracking response.

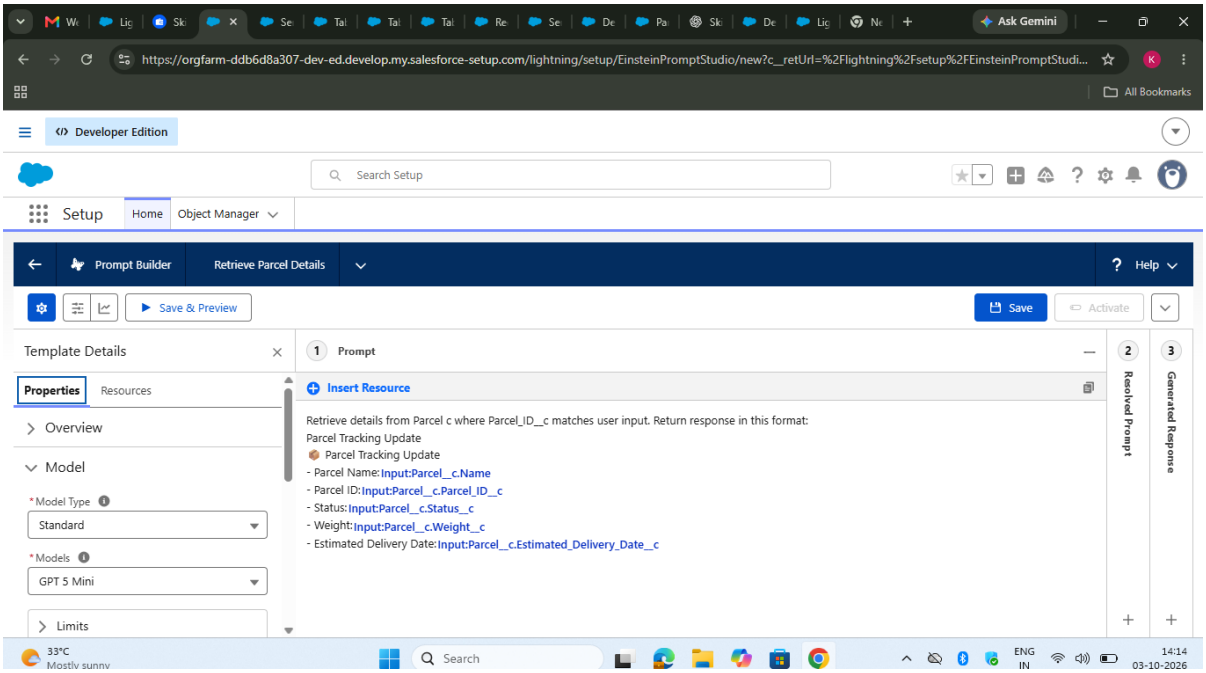


Figure 3. Supplied Salesforce Prompt Builder evidence.

Conclusion

Swift Ship Tracker demonstrates a complete conceptual Salesforce CRM solution for parcel delivery management. The system brings together structured parcel data, operational tracking, automation, AI-assisted information retrieval, customer self-service, security and reporting. The architecture supports the core parcel lifecycle while leaving room for future integrations and advanced logistics features.

The combination of Salesforce configuration and development capabilities makes the project suitable for demonstrating practical CRM skills. Flow provides automation, Apex provides extensibility, Prompt Builder and Agentforce introduce AI-assisted interactions, Experience Cloud supports customer-facing functionality, and reports and dashboards convert operational data into useful management information.

Future Enhancements

- Real-time GPS and map integration.
- SMS and WhatsApp delivery notifications.
- Carrier and payment gateway integration.
- Route optimization and ETA prediction.
- Mobile-first delivery-agent application.
- Advanced AI-based delay detection.
- Automated customer support case creation.
- Historical analytics and predictive delivery insights.

Final Submission Note

For SkillWallet submission, the actual Salesforce org configuration, project stories, testing evidence, GitHub repository and demo link should be verified and attached according to the platform's required workflow. This report is structured as a comprehensive project document and can be used as supporting documentation for the completed implementation.

— END OF PROJECT REPORT —